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Learner: Regional Branch:

Program: Center:

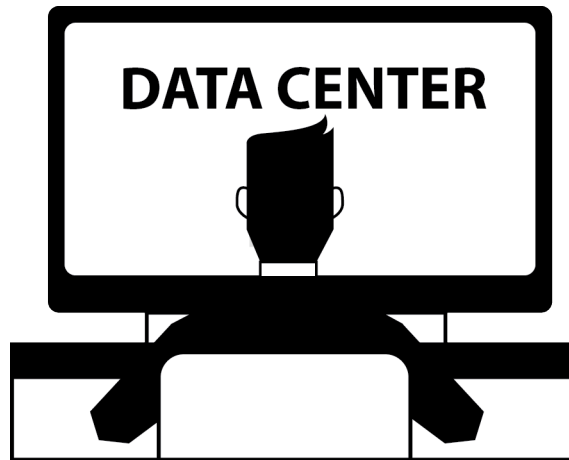
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Let's Start!

Main points of data acquisition



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Understand information about data and its structure in simple spoken and written situations.

Can Do Statements:

- I can give information about data acquisition.
- I can explain different facts related to data acquisition.
- I can analyze how data acquisition works in certain environments.

Ask your partner:

- What devices to collect data from reality do you know? What type of data do they collect?
- What fields do you think are useful in data acquisition? Why?

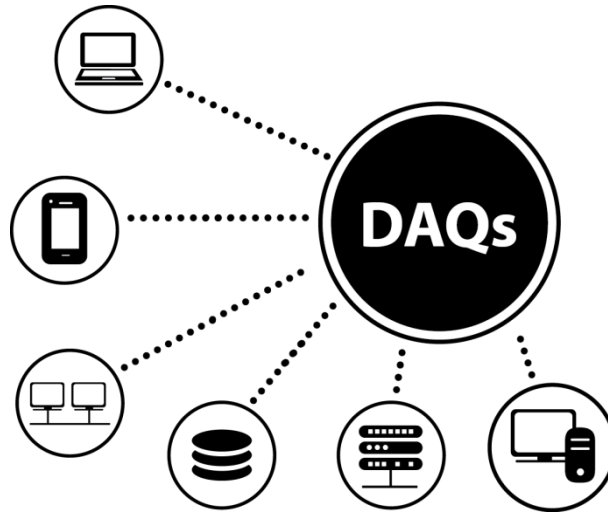




Let's Explore!

Understanding DAQs*

*"DAQs" stands for Data Acquisition devices.



Instructions: Read the text below.

Data acquisition is the process of collecting real-time data (physical phenomena) and digitalize it in real time to manipulate digital representations through the computer. Data acquisition consists of recording facts from reality using a device, (a microphone, a recorder) which sends signals to the computer, it processes the data and presents it on the computer screen. There is a variety of DAQs which turns into different facts concerning the acquisition of data; for example, the keyboard. When we press a key, we are sending a signal which is processed by the computer and finally displayed on the screen as a symbol (a letter or a number) interpreted by the user. When we move the mouse, it sends a signal that allows us to see the cursor on the screen when navigating websites. Digital cameras also convert light levels, when downloading a picture it becomes understandable and easy to interpret as digital images.

There are also some specialized DAQs to measure physical phenomenon and to automatically convert those measurements into digital data to be downloaded. It is represented by using different types of software stored in the memory or in any other hard-copy output device. The datum is usually sent to a computer which uses software to process and display the data using graphs, or other resources, depending on the user's settings.

In certain environments (especially in the field of science and engineering), data acquisition devices help people collect real-time representations. They alert users if there is a change in the data collected in order to adjust any physical fact, as the temperature or light. Also, in the aviation field the usage of automated systems is increasing, as planes functionalities depend on real-time data acquisition.



Let's Practice!

Let's use some important words to talk about data acquisition

Instructions: Match each word with its definition.

COLUMN A

1. Hard copy
2. Navigate websites
3. Send an attachment
4. Download a picture
5. Receive spam

COLUMN B

- a. To transmit a computer file with the email message
- b. A printed version of the data that is in the computer.
- c. To copy pictures from a computer system to another
- d. To obtain junk email or messages that are not from a known sender
- e. The process of moving through resources full of information in a network

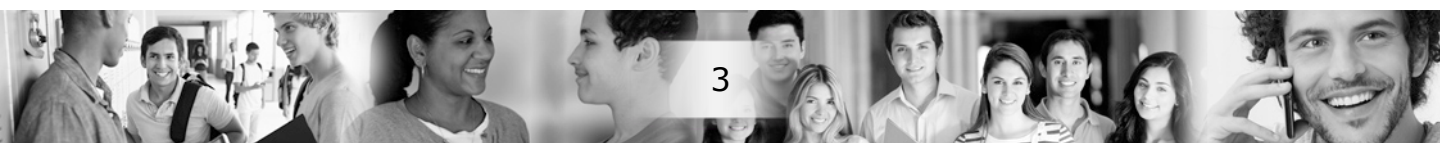


Let's Practice!

Let's check how data acquisition works

Instructions: Read the sentences. Write the missing words in the blank. Use: design, alert users, to manipulate, depend on.

1. Many people _____ data acquisition devices that help people in the field of science and engineering.
2. Data acquisition devices _____ if there is a change in the process of collecting information.
3. Some people use the data collected _____ real time representations.
4. Some functionalities in automated systems _____ real-time data acquisition.





Extension Activity

Now it's your turn to find more information related to data acquisition.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Find information about the sources and systems that data acquisition needs to measure physical phenomenon.
- Choose the most important information.
- Organize the information in a mind map.
- Explain to your partner the information you found about the data acquisition sources and systems.

Self-Assessment

Yes

No

- I identify the sources and systems data acquisition uses to measure physical phenomenon.
- I recognize different facts in data acquisition processes.
- I analyze data in a specific phenomenon.

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Target Words

- **device** (noun): an electronic mechanism
e.g. Data acquisition consist of recording facts from reality using a device.
- **resource** (noun): supply, asset
e.g. The datum is usually sent to a computer which uses software to process and display the data using graphs, or other resources, depending on the user's settings.
- **setting** (noun): in the systems jargon, configuration
e.g. The datum is usually sent to a computer which uses software to process and display the data using graphs, or other resources, depending on the user's settings.
- **screen** (noun): the part of the computer where you can see the information displayed
e.g. It sends a signal that permits us see the cursor on the screen.
- **environment** (noun): factors, in this case, surrounding data
e.g. In certain environments (especially in the field of science and engineering), data acquisition devices help people to collect real-time representations.
- **display** (verb): to show
e.g. The datum is usually sent to a computer which uses software to process and display the data using graphs.



Learning Points

Action Verbs Related to Data Acquisition.

Action verbs are activities people, animals, or things perform. There are some action verbs we use when we talk about data acquisition.

We use many action verbs when we talk about data acquisition, because data acquisition implies the performance of certain actions in order to obtain data.

Examples:

*The DAQ devices **turn** data **into** digital representations.*

*A computer is able to **manipulate** the digital representations.*

The chart below shows some examples of action verbs related to computer science.

Action Verbs Related to Data Acquisition	
Turn into	<i>Devices turn measures into digital forms.</i>
Move	<i>When we move the mouse the cursor receives the information and obeys.</i>
Design	<i>Developers design instruments to collect real-time data.</i>
Convert	<i>Digital cameras convert the light they detect into digital images.</i>
Send	<i>The computer devices send signals that are stored in a memory buffer.</i>
Press	<i>When you press a key, you send a signal to the computer.</i>

Examples

- *When we press a key, we are sending a signal.*
- *When we move the mouse, it sends a signal that permits us to see the cursor on the screen.*
- *Digital cameras also convert light levels, when downloading a picture it becomes understandable and easy to interpret as digital images.*





Learning Points

Articles a/an vs. the, with Countable and Uncountable Nouns.

There are two types of articles in English, definite and indefinite; those articles accompany countable nouns and uncountable nouns.

Indefinite articles are the articles "**a**" or "**an**". The difference between "**a**" and "**an**" is that you use "**a**" with nouns starting with a consonant sound, and you use "**an**" with nouns starting with a vowel sound.

Examples:

A program.

An operation.

The article **a/an** means **one**; for that reason, we only use it with countable singular nouns.

Remember uncountable nouns cannot be count, so there is not one in uncountable nouns, so do not use **a/an** with uncountable nouns.

When you want to talk about nonspecific uncountable nouns, you don't use any article.

Examples:

Inheritance is one of the advantages of abstract data types.

The definite article "**the**" means a specific nouns or nouns, when the speaker and the listener know exactly what we're talking about.

Examples:

The new program

The freelance developers.

We can use the article "**the**" with singular nouns, plural, and uncountable, when we talk about something explicitly and concrete.

Examples:

I bought **the** last version of that software.

We need **the** new software urgently.

I have to abstract **the** datum.

Articles in English		
	Definite	Indefinite
Countable nouns	The records	A record
Uncountable nouns	The software	• software



Learning Points

Examples

- *Data acquisition is the process of collecting real-time data.*
- *Data acquisition consist of recording facts from reality using a device, (a microphone, or a recorder).*
- *It sends a signal that permits us see the cursor on the screen.*
- *The computer processes the data and presents it on the computer screen.*

Common Collocations in Computer Science.

Collocations are groups of words or statements that usually come together.

In the computer science jargon there are several collocations. Let's have a look at some of the most common:

Hard copy: a printed copy of a document.

Download a picture: to save a picture in your computer, you can also download files, music, movies, etc.

Send an attachment: to deliver a document usually via e-mail. **Receive spam:** to get an unpleasant mail usually an advertisement, or a virus.

Navigate websites: to visit sites on the internet.

Examples

- *When we move the mouse, it sends a signal that allows us to see the cursor on the screen when navigating websites.*
- *Digital cameras also convert light levels, when downloading a picture it becomes understandable and easy to interpret as digital images.*
- *It is represented by using different types of software stored in the memory or in any other hard-copy output device.*

